FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – XI
PAPER –2
TEST DATE: 23-08-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. B, D

Sol. $-\frac{dT}{dt} = \frac{e\sigma A}{ms}(T^4 - T_0^4) = \frac{e\sigma A}{\rho VS}(T^4 - T_0^4)$

$$\frac{-\frac{dT_1}{dt}}{-\frac{dT_2}{dt}} = \frac{A_1}{A_2} = \frac{4\pi R^2}{6\ell^2} = \frac{2\pi}{3}\left(\frac{3}{4\pi}\right)^{2/3} = \left(\frac{\pi}{6}\right)^{1/3}$$

Ratio of power radiated by them = $\frac{e\sigma A_1T^4}{e\sigma A_2T^4} = \frac{A_1}{A_2} = \left(\frac{\pi}{6}\right)^{1/3}$

2. A, D

Sol. If n be the principle quantum number of highest energy level after excitation, number of spectral lines in emission spectrum

$= nC_2 = \frac{n(n-1)}{2}$

$\Rightarrow \frac{n(n-1)}{2} = 6 \Rightarrow n = 4$

The energy of the photons which will be responsible for photoelectron emissions are given by

$E_1 = 13.6\left(\frac{1}{1^2} - \frac{1}{4^2}\right) = 12.75$ eV

and $E_1 = 13.6\left(\frac{1}{1^2} - \frac{1}{3^2}\right) = 12.09$ eV

The maximum kinetic energies of the photoelectrons emitted in the two cases are

$k_1 = h\nu - \phi = 12.75 - \phi$

and $k_2 = h\nu - \phi = 12.09 - \phi$

Given, $k_1 = (1.1)k_2 \Rightarrow 12.75 - \phi = 1.1(12.09 - \phi)$

$\phi = 5.48$ eV

3. A, B, C

Sol. $\frac{|Q_1|}{2\varepsilon_0}(1-\cos 37°) = \frac{|Q_2|}{2\varepsilon_0}(1-\cos 60°) \Rightarrow \frac{|Q_1|}{|Q_2|} = \frac{5}{2}$

Let potential at point p(x, y) is zero in xy-plane. Then,

$$\frac{kQ_1}{\sqrt{x^2+y^2}} + \frac{-kQ_2}{\sqrt{(x-a)^2+y^2}} = 0$$

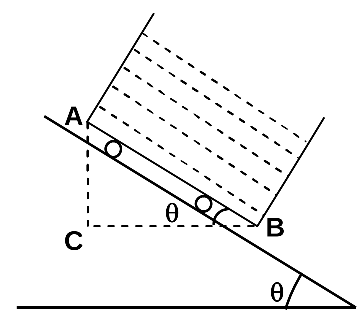
$$\frac{(x-a)^2+y^2}{x^2+y^2} = \frac{(Q_2)^2}{(Q_1)^2} = \left(\frac{2}{5}\right)^2$$

$$\Rightarrow x^2+y^2-\frac{50ax}{21}+\frac{25a^2}{21} = 0$$

So, the centre of circle is $\left(\frac{25a}{21}, 0\right)$ and radius of circle is $\frac{10a}{21}$

4. B, C

Sol. If the free surface of liquid makes an angle α with the horizontal, then



$$\tan\alpha = \frac{a\cos\theta}{g-a\sin\theta} = \frac{\frac{g}{n}\cos\theta}{g-\frac{g}{n}\sin\theta} = \frac{\cos\theta}{n-\sin\theta}$$

$$\alpha = \tan^{-1}\left(\frac{\cos\theta}{n-\sin\theta}\right)$$

$$P_C - P_A = \rho g\left(1-\frac{\sin\theta}{n}\right)\ell\sin\theta$$

$$P_C - P_B = \rho\frac{g}{n}\cos\theta\,\ell\cos\theta$$

$$P_B - P_A = \frac{\rho g\ell}{n}\left[\cos^2\theta-(n-1)\sin^2\theta\right]$$

$$= \rho g\ell\left(\sin\theta-\frac{1}{n}\right)$$

5. B, C, D

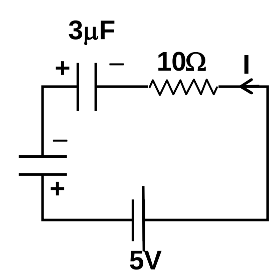
Sol. Let I be the current just after closing the switch

Then from loop rule

$5 - I \times 10 + 2 + 3 = 0$

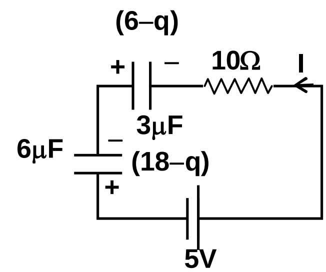
$\Rightarrow I = 1A$

Let q charge flow through the circuit in anticlockwise sense after the switch is closed.



Then $5+\left(\frac{6-q}{3}\right)+\left(\frac{18-q}{6}\right) = 0$

$\Rightarrow q = 20\ \mu C$



Work done by the battery
$W_b = \varepsilon q = 5 \times 20 = 100\ \mu J$
Final potential difference on the 3μF capacitor
$= \frac{6-q}{3} = \frac{6-20}{3} = -\frac{14}{3}V$
Final potential difference on 6μF capacitor
$= \frac{18-q}{6} = \frac{18-20}{6} = -\frac{1}{3}V$
Negative sign of potential difference on capacitors implies that polarity of the plates are opposite to as shown in the figure.
Change in energy of capacitor
$= U_2 - U_1 = \left\{\frac{1}{2}\times 3\times\left(\frac{14}{3}\right)^2 + \frac{1}{2}\times 6\times\left(\frac{1}{3}\right)^2\right\} - \left\{\frac{1}{2}\times 3\times 2^2 + \frac{1}{2}\times 6\times 3^2\right\} = 0$
Heat produced = $W_b - \Delta U = 100\ \mu J$

6. A, B
Sol. At the particles reach point P at same time
$\frac{2v_0 \sin\alpha}{g} = \frac{v_0}{g\sin\beta} \Rightarrow 2\sin\alpha = \frac{1}{\sin\beta}$
Since their horizontal displacement are equal
$v_0 \cos\alpha t = \frac{v_0}{2} t\cos\beta$
$2\cos\alpha = \cos\beta$

7. B
Sol. Consider an element of ring of radius r and thickness dr
Resistance of this small element dR = $\rho\frac{2\pi r}{hdr}$
Induced emf in this element $\varepsilon = \left|\frac{d\phi}{dt}\right| = \pi r^2 \frac{dB}{dt} = \pi r^2 k$
Current in this element $dI = \frac{d\varepsilon}{dR} = \pi r^2 k \frac{hdr}{\rho 2\pi r} = \frac{khrdr}{2\rho}$
Current in the entire ring
$I = \int dI = \frac{kh}{2\rho}\int_a^b rdr = \frac{kh(b^2 - a^2)}{4\rho}$

8. D

9. A

10. C
Sol. **(for Q. 9 to 10):**
The vertical component of velocity of the two pieces will not change due to explosion as they reach the ground simultaneously. Let the piece which return to origin hits the ground at an angle θ with horizontal.
Then $20\sqrt{2}\sin\theta = 40\sin 30° \Rightarrow \theta = 45°$.
If the piece when return to origin would have been projected from origin O with speed $20\sqrt{2}$ m/s at angle 45° with horizontal it would pass through the point of explosion.

As they follow the same trajectory

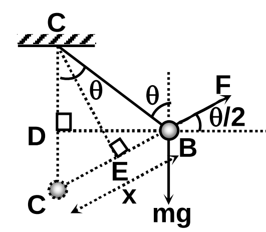
$$x\tan 30^\circ - \frac{gx^2}{2(40)^2\cos^2 30^\circ} = x\tan 45^\circ - \frac{gx^2}{2\left(20\sqrt{3}\right)^2\cos^2 45^\circ}$$

$$\Rightarrow\ x = 120\left(1-\frac{1}{\sqrt{3}}\right)\text{m} = 50.76\text{ m}$$

Section – B

11. 3

Sol. For similar triangles ABD and ACE



$$\frac{x/2}{\ell} = \frac{h}{x} \quad \text{…(i)}$$

For equilibrium of ball B

$$\frac{mg}{\sin\left(90+\frac{\theta}{2}\right)} = \frac{F}{\sin(\pi-\theta)}$$

$$\Rightarrow\ \frac{mg}{kq^2/x^2} = \frac{\cos\left(\frac{\theta}{2}\right)}{\sin\theta} = \frac{1}{2\sin\left(\frac{\theta}{2}\right)} = \frac{1}{2\frac{h}{x}}$$

$$\Rightarrow\ \frac{kq^2}{x} = 2mgh$$

$$W = \Delta U_{\text{electrostatic}} + \Delta U_{\text{gravitational}}$$

$= 2mgh + mgh = 3mgh = 3 \times 1 \times 10 \times 0.1 = 3$ J

12. 262

Sol. For refraction at first surface

$$\frac{\mu}{v_1} - \frac{1}{-\infty} = \frac{\mu-1}{R}$$

$$\Rightarrow\ v_1 = \frac{\mu R}{\mu-1}$$

For second refracting surface

$$u_2 = v_1 - R = \frac{\mu R}{\mu-1} - R = \frac{R}{\mu-1}$$

For refraction at second surface

$$1-\mu(\mu-1) = 2-2\mu$$

$$\Rightarrow\ \mu^2 - 3\mu + 1 = 0$$

$$\mu = \frac{3\pm\sqrt{5}}{2}$$

Since, $\mu = \frac{3-\sqrt{5}}{2}$ is not acceptable

$$\mu = \frac{3+\sqrt{5}}{2} = \frac{3+2.24}{2} = 2.62$$

13. 3

Sol. $\frac{1+1}{\gamma_{min}-1} = \frac{1}{\frac{5}{3}-1} + \frac{1}{\frac{7}{5}-1} \Rightarrow \gamma_{min} = \frac{3}{2}$

$M_{mix} = \frac{n_1M_1 + n_2M_2}{n_1+n_2} = \frac{1\times 4 + 1\times 32}{2} = 18$

$f_1 = \frac{3v_1}{4\ell_1},\ f_2 = \frac{3v_2}{2\ell_2}$

$\frac{f_1}{f_2} = \frac{6v_1}{v_2}\frac{\ell_2}{2\ell_1} = 6\sqrt{\frac{\gamma_{H_2}}{\gamma_{mix}}\frac{T_{H_2}}{T_{mix}}\frac{M_{mix}}{M_{H_2}}} \times 2 = 3$

Section – C

14. 74.79

15. 340.00

Sol. **(for Q. 14 to 15):**

$W_{AB} + W_{BC} + W_{CA} = W_{CD} + W_{DA} + W_{AC}$

$0 + W_{BC} - W_{AC} = 0 + W_{DA} + W_{AC}$

$W_{BC} - W_{DA} = 2W_{AC}$

$W_{BC} = R(T_C - T_B) = R(400 - T_B)$

$W_{DA} = R(T_A - T_D) = R(289 - T_D)$

A$\rightarrow$C is a polytropic process $(P \propto V \Rightarrow PV^{-1} =$ constant)

For process $A \rightarrow C$ $W_{AC} = \frac{R}{(1-\alpha)}(T_C - T_A) = \frac{R}{2}(T_C - T_A)$

$\Rightarrow R(400 - T_B) - R(289 - T_D) = \frac{2R}{2}(400 - 289)$

$111 - T_B + T_D = 111$

$\Rightarrow T_B = T_D$

For process $A \rightarrow B$

$\frac{P_1}{T_A} = \frac{P_2}{T_B} \Rightarrow T_B = \frac{P_2}{P_1}289$ …(i)

$\frac{P_2}{T_C} = \frac{P_1}{T_D} \Rightarrow T_D = \frac{P_1}{P_2}400$ …(ii)

From (i) and (ii)

$T_BT_D = 289 \times 400$

$\Rightarrow T_B^2 = 289 \times 400$

$T_B = 17 \times 20 = 340$ K

$W_{AB} = 0$, $W_{BC} = R(T_C - T_B) = R(400 - 340) = 60R$

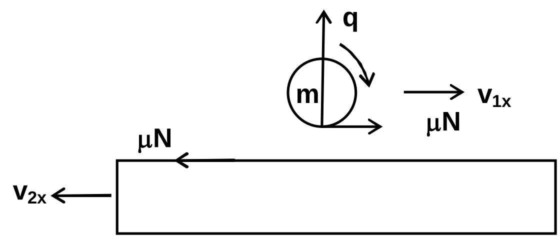
$W_{CD} = 0$, $W_{DA} = R(T_A - T_D) = R(289 - 340) = -51R$

$W_{net} = W_{AB} + W_{BC} + W_{CD} + W_{DA} = 9R = 9 \times 8.31 = 74.79$ J

16. 06.00

17. 25.00

Sol. **(for Q. 16 to 17):**



$N\Delta t = 2mv_0$

$\mu N\Delta t = mv_{1x}$

$\Rightarrow \mu 2mv_0 = mv_1x$

$\Rightarrow v_{1x} = 2\mu_0 v_0 = 2\times 0.1\times\sqrt{2\times 10\times 5} = 2\,\text{m/s}$

$mv_{1x} - Mv_{2x} = 0$

$\Rightarrow v_{2x} = \frac{1\times 2}{2} = 1\text{m/s}$

Distance CD

$= (v_{1x} + v_{2x})\frac{2v_0}{g} = 3\times\frac{2\times\sqrt{2\times 10\times 5}}{10} = 6\text{ m}$

For sphere

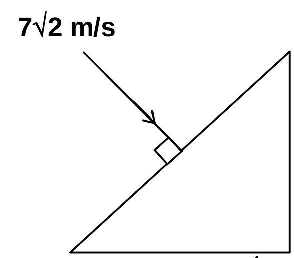
Angular impulse = $I\Delta\omega$

$\Rightarrow \mu NR\Delta t = \frac{2}{5}mR^2\Delta\omega \Rightarrow \Delta\omega = \frac{5\mu mv_0}{MR} = 25\ \text{rad/s}$

18. 04.00

19. 12.00

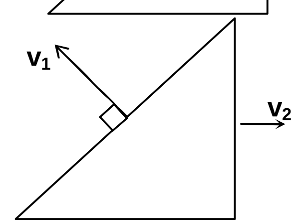
Sol. **(for Q. 18 to 19):**



The vertical component of velocity of ball just before collision

$v_y = \sqrt{2a_y y}$

$= \sqrt{2\times 9.8\times 2.5} = 7\ \text{m/s}$



If v_2 and v_1 be the velocit of wedge and the ball just after collision, conserving momentum of the system along horizontal direction

$3v_2 - 1\times v_1\cos 45° = 1\times 7$ …(i)

Also, $\frac{v_2\cos 45° - (-v_1)}{7\sqrt{2} - 0} = 1$ …(ii)

From equation (i) and (ii), v_2 = 4 m/s and $v_1 = 5\sqrt{2}$ m/s

Impulse exerted by the wedge on the ball

$\Delta p = 1\times 5\sqrt{2} - 1\times(-7\sqrt{2}) = 12\sqrt{2}$ kg-m/s

Vertical component of impulse = $12\sqrt{2}\cos 45° = 12$ N-s

Chemistry

PART – II

Section – A

20. A, C

Sol. Radial node $= n - \ell - 1$

$= 2 - 0 - 1$

$= 1$

$\Rightarrow \psi_R$ for 2s $= \left(\frac{1}{2\sqrt{2}}\right)\left(\frac{Z}{a_o}\right)^{3/2}\left(2 - \frac{Zr}{a_o}\right)e^{-Zr/2a_o}$

$\Rightarrow$ At radial node $2 - \frac{Zr}{a_o} = 0$

Hence, $\frac{Zr}{a_o} = 2$

$\Rightarrow$ Angular function for $s = \left(\frac{1}{4\pi}\right)^{1/2}$

21. A, B, D

Sol. MO configurations

$N_2 = KK\ \sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x{}^2 = \pi 2p_y{}^2\ \sigma 2p_z{}^2$

$F_2 = KK\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_z{}^2\ \pi 2p_x{}^2 = \pi 2p_y{}^2\ \pi^* 2p_x{}^2 = \pi^* 2p_y{}^2$

O_2^{--} = same as F_2

$C_2 = KK\ \sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^2 = \pi 2p_y^2$

22. C, D

Sol. Transuranium : The elements after uranium in periodic table are called Transuranium elements.

23. C, D

Sol. The number of cation vacancy will be equal to twice the number of $FeCl_3$.

Number of cation vacancy per mole of NaCl $= \frac{10^{-3} \times 6.02 \times 10^{23} \times 2}{100}$

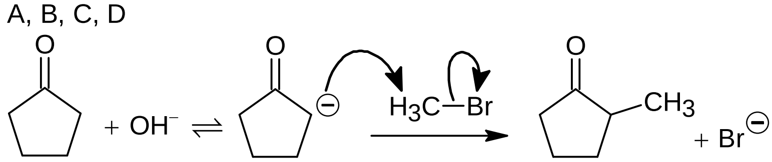
$= 12.04 \times 10^{18}$

24. A, B, C, D

Sol. $-CH_3$ is an activating group and ortho para director.

25. A, B, C, D

Sol.



Similarly multialkylation will be done.

26. D

Sol. $\lambda^o_{H^+} = 349.6\ Scm^2\ mol^{-1}$

$\lambda^o_{OH^-} = 199.1\ Scm^2\ mol^{-1}$

$\lambda^o_{Na^+} = 50.1\ Scm^2\ mol^{-1}$

27. B

Sol. $\Lambda_m = \kappa \times \dfrac{1000}{M}$

$= 4.95 \times 10^{-5} \times \dfrac{1000}{0.001028} = 48.15\ \text{Scm}^2\ \text{mol}^{-1}$

$\alpha = \dfrac{\Lambda_m}{\Lambda_m^o} = \dfrac{48.15}{390.15} = 0.1234$

$K_a = \dfrac{C\alpha^2}{(1-\alpha)} = \dfrac{0.001028 \times (0.1234)^2}{0.8766}$

$K_a = 1.78 \times 10^{-5}$

28. C

Sol.

$$\underset{(A)}{RCH_2OH} \xrightarrow[\text{Pyridine}]{PBr_3} \underset{(B)}{R - CH_2Br}$$

$$\underset{(A)}{RCH_2OH} \xrightarrow{NaOCl,\ CH_3COOH,\ 0^oC} \underset{(D)}{RCHO}$$

$$\underset{(B)}{R - CH_2Br} \xrightarrow{Mg,\ Et_2O} \underset{(C)}{RCH_2MgBr}$$

$$\underset{(D)}{RCHO} + \underset{(C)}{RCH_2MgBr} \longrightarrow \underset{(E)}{R-\overset{\overset{\large OMgBr}{|}}{\underset{\underset{\large H}{|}}{C}}-CH_2R} \xrightarrow{H_3O^+} \underset{\underset{(2^o\ \text{alcohol})}{(F)}}{R-\overset{\overset{\large OH}{|}}{C}H-CH_2R}$$

29. B

Sol.

$$\underset{(D)}{R-\underset{\underset{\large H}{|}}{C}=O} + CN^- (\text{excess}) \xrightarrow{HCl} \underset{(H)}{R-\overset{\overset{\large OH}{|}}{\underset{\underset{\large H}{|}}{C}}-CN} \xrightarrow[\Delta]{HCl,\ H_2O} \underset{(I)}{R-\overset{\overset{\large OH}{|}}{C}H-COOH}$$

Section – B

30. 1500

Sol. $\dfrac{[B]}{[C]} = \dfrac{5}{4}$ $\qquad \dfrac{[C]}{[D]} = \dfrac{4}{1}$ $\qquad \dfrac{[B]}{[D]} = \dfrac{5}{1}$

$[B] : [C] : [D] = 5 : 4 : 1$

$k_1 = 5x\ \ k_2 = 4x\ k_3 = x$

$k = \dfrac{0.693}{150} = 0.00462$

$k = k_1 + k_2 + k_3$

$0.00462 = 5x + 4x + x$

$x = \dfrac{0.00462}{10} = 0.000462$

$$t_{1/2}\left(\text{partial half life}\right) = \frac{0.693}{k_3} = \frac{0.693}{0.000462}$$

= 1500 hrs.

31. 180

Sol. $\Delta S_{lake} = \frac{q_{irr}}{T} = \frac{mS\Delta T}{T} = \frac{566 \times 90}{283}$

= 180

32. 2

Sol. $KCl + AgNO_3 \longrightarrow AgCl + KNO_3$

$M_1V_1 = M_2V_2$

$M_1 = \frac{M_2V_2}{V_1}$

$M_1 = \frac{1 \times 20}{36} = \frac{20}{36}$

$M_1 = \text{molality} = \frac{20}{36}$

$i = 1 + \alpha(n-1)$

$i = 1 + \alpha$

$= 1 + 0.8$

= 1.8

$\Delta T_f = iK_f m$

$= 1.8 \times 2 \times \frac{20}{36}$

= 2

Section – C

33. 4000.00

Sol. $ee = \frac{6-4}{6+4} \times 100 = \frac{2}{10} \times 100 = 20$

$x = \frac{20}{0.005} = 4000$

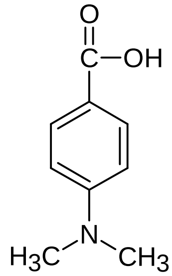
34. 9000.00

Sol. $y = \frac{20 \times 13.5^{o}}{100 \times 0.0003}$

$y = 9000$

35. 5000.00

Sol.

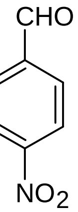


degree of unsaturation = 5.

$x = 5 \times 10^{3}$

36. 3775.00

Sol.



$$y = \frac{M.wt. \times x}{200} = \frac{151 \times 5000}{200} = 3775$$

37. 2370.00

Sol.

$$K\overset{+5}{I}O_3 + K\overset{-1}{I}(\text{excess}) \longrightarrow \overset{0}{I}_2$$

$$n_f = 5 \quad n_f = 1 \quad n_f = \frac{5}{3}$$

Moles of I_2 = moles of $KIO_3 \times 3$
= 10 × 3 = 30
Milliequivalents of I_2 = moles of $n_f \times 1000$

$$= 30 \times \frac{5}{3} \times 1000$$

$$= 50 \times 1000$$

$$I_2 + 2Na_2S_2O_3 \longrightarrow Na_2S_4O_6 + 2NaI$$

$$n_f = 2 \quad n_f = 1$$

$$N_1V_1 = N_2V_2$$

$$N_2 \text{ of } Na_2S_2O_3 = \frac{N_1V_1 \text{ of } I_2}{V_2}$$

$$= \frac{50000 \times 6}{5 \times 20 \times 1000}$$

= 3

Strength of $Na_2S_2O_3 = N \times E$

$$= \frac{3 \times 158}{1} = 474 \text{ g}/\ell$$

$$x = \frac{S_{hypo} \times 10}{2}$$

$$= 2370$$

38. 2500.00

Sol. $y = 60000 \times \frac{5}{3} \times \frac{1}{40}$

$$y = 2500$$

Mathematics PART – III

Section – A

39. A, B, C, D

Sol. $\alpha = \sum_{k=1}^{\infty} \tan^{-1}\left(\sqrt{k+2} - \tan^{-1}\sqrt{k}\right) = \frac{3\pi}{4} - \tan^{-1}\sqrt{2}$

40. B, D

Sol. $$f(x) = \begin{cases} 2\pi + 4\tan^{-1}\frac{x}{3} & ; \quad x < -3 \\ \pi & ; \quad -3 \le x < 0 \\ \pi - 4\tan^{-1}\frac{x}{3} & ; \quad 0 \le x < 3 \\ 0 & ; \quad x \ge 3 \end{cases}$$

41. B, C

Sol. $A^{-1} = A^T - A\cdot\text{Adj}\ (2B^{-1}))$; $AA^{-1} = AA^T - A^2\ \text{Adj}\ (2B^{-1})$

$\Rightarrow A^2\ \text{Adj}\ (2B^{-1}) = AA^T - I = 6I$

$|A|^2\ |\text{Adj}(2B^{-1})| = |6I| \Rightarrow \text{Det Adj}\left(2B^{-1}\right) = \frac{36}{49}$

Also, $|A|^2 \times \frac{4}{|B|} = 36 \Rightarrow |B| = \frac{49\times 4}{36} = \frac{49}{9}$ and $\text{Det}\left(B^{-1}\right) = \frac{9}{49}$

42. A, B, C, D

Sol. $3\ell - m + 4n = 3$; $l + 2m - 3n = -2$; $6\ell + 5m - 5n = -3$

Given equation have infinite solution given by $\ell = \frac{4-5n}{7}; m = \frac{13n-9}{7}; n \in R$

Equation of plane containing L_1 and L_2 is $3x + 4y - 5z + 4 = 0$ ….. (π)

$\because$ Point P(l, m, n) satisfy the plane π

$\therefore$ l = 2, m = –5, n = –2

43. A, B, C

Sol. $f'(c) = 0$; $c \in (1, 6)$. Using LMVT for $f'(x)$ in [1, c] and [c, 6]

$|f''c_1| = \left|\frac{f'(c) - f'(1)}{c-1}\right| \le 10$ ….. (1)

$|f''c_2| = \left|\frac{f'(6) - f'(c)}{|6-c|}\right| \le 10$ ….. (2)

From equation (1) and (2), we get $|f'(c) - f'(1)| \le 10(c-1)$ and $|f'(6) - f'(c)| \le 10(6-c)$

$\Rightarrow |f'(1)| \le 10(c-1)$ and $|f'(6)| \le 60 - 10c$

$\therefore |f'(1)| + f'(6)| \le 50$

44. A, B, C, D

Sol. $\because$ $a(1-e) = 5 \Rightarrow a = 20$

$\therefore$ $b^2 = 400\left(1 - \frac{9}{16}\right) = 175$

$\because$ $P_1P_2 = b^2$ $\therefore$ $P_1 \times 5 = 175 \Rightarrow P_1 = 35$

45. B

46. A

Sol. **(for Q. 45 to 46):**

$f(1)=\left|\sin(3a-1)\frac{\pi}{4}\right|-\left|\cos\frac{b\pi}{4}\right| \Rightarrow (3a-1)\frac{\pi}{4}=\frac{\pi}{2},\frac{5\pi}{4},\frac{8\pi}{4},\frac{11\pi}{4},\frac{14\pi}{4},\frac{17\pi}{4}$

and $b\frac{\pi}{4}=\frac{\pi}{4},\frac{2\pi}{4},\frac{3\pi}{4},\frac{4\pi}{4},\frac{5\pi}{4},\frac{6\pi}{4}$ $\therefore$ f(1) = 0 or 1

$\therefore$ P(E) = (2× 3 + 3 × 3) / 6 × 6 = $\frac{5}{12}$; $f\left(\frac{2}{3}\right)=\left|\sin(3a-1)\frac{\pi}{6}\right|-\left|\cos\frac{b\pi}{6}\right|$

$(3a-1)\frac{\pi}{6}=\frac{2\pi}{6},\frac{5\pi}{6},\frac{8\pi}{6},\frac{11\pi}{6},\frac{14\pi}{6},\frac{17\pi}{6}$; $\frac{b\pi}{6}=\frac{\pi}{6},\frac{2\pi}{6},\frac{3\pi}{6},\frac{4\pi}{6},\frac{5\pi}{6},\frac{6\pi}{6}$

$f\left(\frac{2}{3}\right)=\frac{1}{2}-\frac{\sqrt{3}}{2}$ or $\frac{\sqrt{3}}{2}-\frac{1}{2},\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}-1$; $P(E)=\frac{3\times2+3\times4}{6\times6}=\frac{1}{2}$

47. C

48. A

Sol. $a_1 + 4d = 31$ and $2a_1 + 5d = P_2$; prime

$\Rightarrow$ Possible cases $a_1 = 3, d = 7$; $a_1 = 11, d = 5$; $a_1 = 19, d = 3$

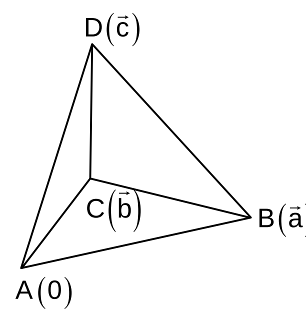
Section – B

49. 10

Sol. Shortest distance between edges AC and BD is given by

$$k=\frac{\left|\vec{a}\cdot\left(\vec{b}\times\vec{c}+\vec{a}\times\vec{b}\right)\right|}{\left|\vec{b}\times\vec{c}+\vec{a}\times\vec{b}\right|}$$

$$k=\frac{\left[\vec{a}\ \vec{b}\ \vec{c}\right]}{\left|\vec{b}\right|\left|\vec{c}-\vec{a}\right|\sin\theta}=10$$



50. 5

Sol. Rotation about A

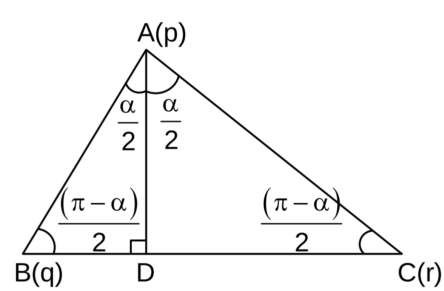
$$\frac{r-p}{q-p}=e^{i\alpha} \quad \text{..... (1)}$$

Rotation about B

$$\frac{p-q}{r-q}=\frac{AB}{BC}e^{\frac{i(\pi-\alpha)}{2}}=\frac{AB}{2BD}e^{i\left(\frac{\pi-\alpha}{2}\right)} \quad \text{..... (2)}$$

$$\frac{r-p}{q-p}\left(\frac{p-q}{r-q}\right)^2=\frac{1}{4}\left(\frac{AB}{BD}\right)^2e^{i\pi}$$

$\Rightarrow (q-r)^2 = 4(p-q)(r-p)\sin^2\frac{\alpha}{2}$ $\therefore$ $\alpha=\frac{5\pi}{6}$



51. 25

Sol. Put $2x^{15}-3x^{10}+6x^5=t$

Section – C

52. 04.00

53. 208.00

Sol. **(for Q. 52 to 53):**

$\because$ $II_1 = 20\sqrt{2}$, R = 10 $\Rightarrow \angle A = 90°$. Also a = 2R sin A = 20

$r = (s-a)\tan\frac{A}{2} = 4$.

Also, length of median through vertex C is $\frac{\sqrt{2(a^2+b^2)-c^2}}{2}$

54. 7884.00

55. 04.00

Sol. **(for Q. 54 to 55):**

Total number of matrices in S are $\frac{9!}{7!}+\frac{9!}{6!}+\frac{9!}{4!3!}+\frac{9!}{6!2!}+\frac{9!}{5!2!2!}+\frac{9!}{4!2!2!} = 7884$

56. 33.00

57. 229.00

Sol. $g'(f(x)) = \frac{1}{f'(x)} \Rightarrow g'(109) = \frac{1}{f'(4)}$